

Public Economics (ECON 131)

Section #8: Externalities (EXTRA)

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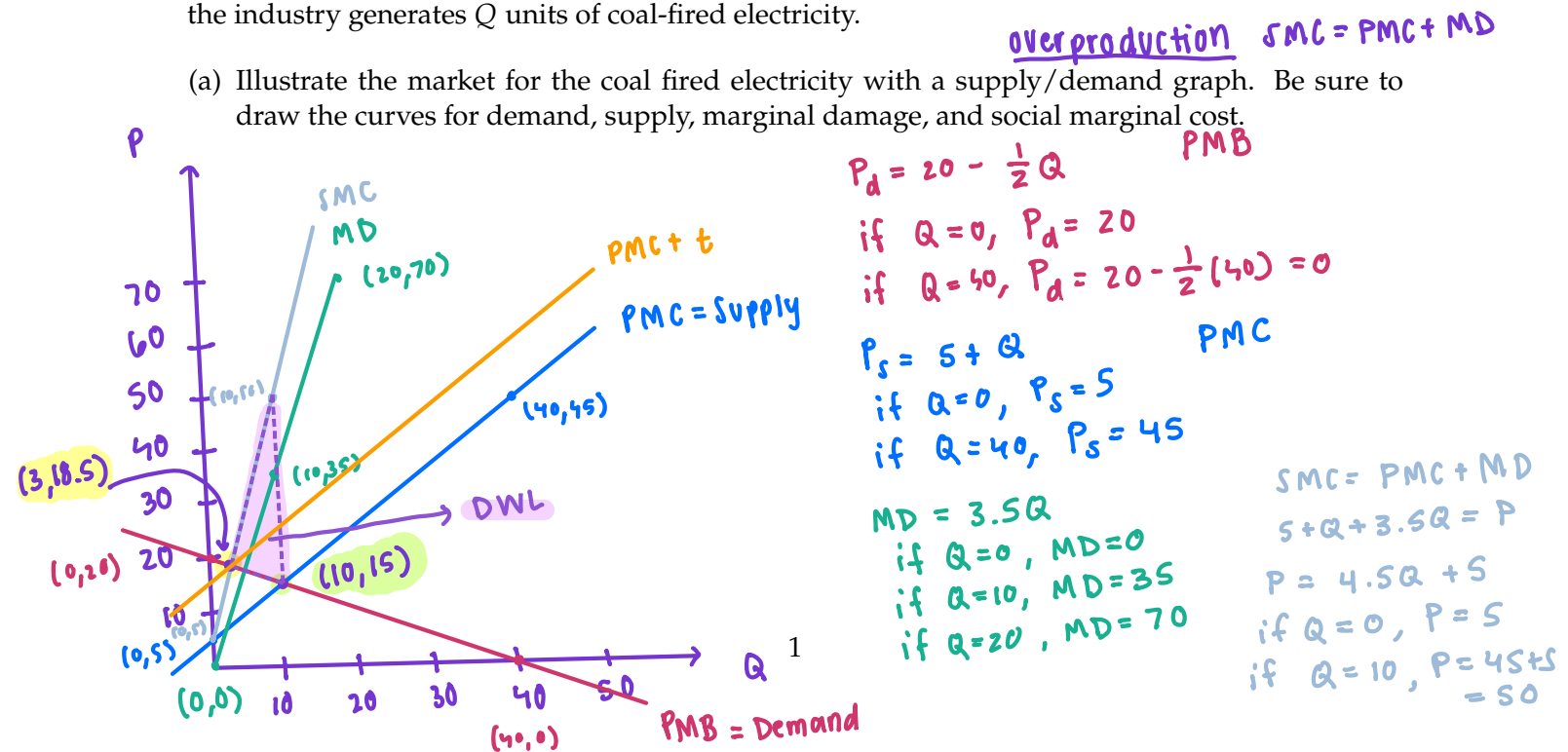
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1 Externalities Cont.

1.1 When Marginal Damage is Not Constant

A coal-fired power plant releases air pollution into the atmosphere for every unit of electricity produced. The inverse demand function for coal-fired electricity is $P_d = 20 - \frac{1}{2}Q$, which represents the marginal benefit curve where Q is the quantity consumed when consumers pay price P_d . The inverse supply curve for coal-fired electricity is $P_s = 5 + Q$, which represents the marginal private cost curve when the power plant produces Q units. The marginal damage from emissions is given by $MD = 3.5Q$, which describes the cost of greenhouse gas emissions and local air pollution when the industry generates Q units of coal-fired electricity.

- (a) Illustrate the market for the coal fired electricity with a supply/demand graph. Be sure to draw the curves for demand, supply, marginal damage, and social marginal cost.



- (b) What are the equilibrium price and quantity for coal fired electricity when there is no correction for the externality?

asking for free market Q and P.

set, $PMC = PMB$ s.t.

$$5 + Q = 20 - \frac{1}{2}Q$$

$$Q + \frac{1}{2}Q = 20 - 5$$

$$\frac{3}{2}Q = 15$$

$$Q_{FM} = 15 \cdot \frac{2}{3} = 10$$

$$P = 5 + Q_{FM}$$

$$P_{FM} = 5 + 10 = 15$$

- (c) How much coal fired electricity should the market supply at the social optimum?

$$SMC = SMB$$

here, overproduction (air pollution from steel plant)

$$\Rightarrow PMC + MD = PMB$$

$$5 + Q + 3.5Q = 20 - \frac{1}{2}Q$$

$$4.5Q + 0.5Q = 20 - 5$$

$$5Q = 15$$

$$Q_{SO} = 3$$

$$P_{SO} = 20 - \frac{1}{2}Q_{SO}$$

$$= 20 - \frac{1}{2}(3)$$

$$= \frac{40}{2} - \frac{3}{2} = \frac{37}{2}$$

$$P_{SO} = 18.5$$

- (d) How large is the deadweight loss from the externality?

$$DWL = \frac{1}{2} \cdot b \cdot h$$

the height (h) is the $Q_{FM} - Q_{SO} = 10 - 3 = 7 = h$

the base (b) is MD when Q_{FM} , $MD = 3.5Q_{FM}$
 $= 3.5(10)$
 $= 35$
 $= b$

$$DWL = \frac{1}{2} \cdot 35 \cdot 7$$

$$DWL = 122.5$$

- (e) Is it possible for the government to achieve the social optimum by imposing a per-unit fee on emissions? If not, explain why it is not possible. If so, how large must the emission fee be if the market is to produce the socially efficient amount of coal fired electricity? Also, draw the firm's supply curve with the new emission fee on your graph.

yes, we ultimately want the tax to be equal to the MD.

$$\text{so, } t^* = MD = 3.5Q$$

the Q is the socially optimum quantity

$$\text{so, } t^* = 3.5Q_{so} = 3.5 \times 3 = 10.5$$

therefore our new supply curve w/ unit tax is

$$SMC = P_s = PMC + MD = 5 + Q + t = 5 + Q + 10.5$$

$$P_s = 15.5 + Q$$

in other words,
the tax is equal
to the amount of
marginal damage
at the socially
optimum quantity